# On The Gravitational Nature Of Time

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**Abstract**

We explore a modified set of coordinate transformations that describe time, cosmic inflation and gravitational acceleration as the same process as observed from different reference frames through an unwrapped Kaluza-Klein like geometry. These transformations satisfy existing experimental validations of special relativity while providing a set of 4 equations yielding our peculiar velocity, a resolution to the SNe/CMB velocity discrepancy, an orbital velocity equation that matches observation with only visible mass, and a direct physical mechanism to the equivalence principle.

**Keywords:** gravitation, time, relativity, inflation

## 1 Introduction

Albert Einstein was insistent on a static Universe during the period in which he wrote his specialized theory of relativity[1–4](SR). What is proposed here is largely a re-examination of this geometry under the assumption that the $\mathbb{R}^3$ space between two arbitrary points dilates as time progresses.

This dilation is then bound to $\gamma$, giving a velocity dependent function for $\vec{g}$ and a magnitude that extends into a 4th spatial coordinate, similar in construction to an unwrapped Kaluza-Klein (KK) dimension[5, 6]. Through these coordinate transformations time, cosmic inflation and gravitational acceleration become the same process as observed from different reference frames in a space-time where the 4th KK dimension acts as a concurrent density axis.

This geometry gives an alternative model of concurrency and synchronicity as a classical notion of simultaneity is maintained. While maintaining classical simultaneity resolves numerous temporal paradoxes without the need for selective reference frame

preference, it produces a concurrent axis that describes the ultra-dense state of the early Universe as being separated only by some $\Delta$ along this 4th axis.

### 1.1 The Lighthouse & The Clock-Tower

Let an event be defined classically, such that a single, instantaneous event exists at a single point in space-time; a local measurement as $dt \to 0$. The constant nature of 4-vecs inherent to the definition of an event in SR arises naturally from the coordinate transformations that follow, giving $d/v$ that is equivalent as measured in each respective coordinate system.

Let there be two overlapping coordinate systems, **S** and **U**. Let **S** and **U** represent static positions in time, with **U** representing a 'future' state from the perspective of an observer in **S** so that

$$v^{(S)} = (t, x_\mu) \qquad v^{(U)} = (t + \Delta t, x_\mu) \tag{1}$$

where $\Delta t > 0$.

Let there also be two observers, $A$ and $B$. Let $A$ and $B$ agree that $B$ should travel at some relativistic velocity $v'$ between two arbitrary points in space contained within **S** and **U**, $p_1$ and $p_2$. Let $A$ remain at relative rest with respect to $p_1$, and at $p_2$ place a time keeping device large enough to be visible from $p_1$.

If $A$ and $B$ were to disagree upon the time at which $B$ reaches $p_2$, this single, instantaneous event ($B$ reaching $p_2$) would be required to happen more than once. Therefore, we theorize that it is not possible for $A$ and $B$ to disagree upon the time at which $B$ reaches $p_2$.

Instead, as the additional factor of $\gamma$ has been demonstrated experimentally on a number of occasions[7–11], we should conclude that both $A$ and $B$ observe $B$ reaching $p_2$ in $\frac{d}{v}\gamma$. This gives a set of coordinate transformations such that for each respective observer

$$\Delta t = \frac{\overbrace{d \cdot \gamma}^{B}}{v^{(U)}} = \underbrace{\frac{d}{v^{(S)}} = \frac{d}{v^{(U)}\left(\frac{1}{\gamma}\right)}}_{A} \tag{2}$$

This produces a model of relativistic dilation that loosely binds the additional $\gamma$ factor in the $d = vt\gamma$ equivalence to cosmic inflation, where the additional velocity experienced by $B$ results in a more rapid progression along this concurrent density axis leading to a further separation between any two points in **U**.

## 2 Theoretical Framework

Let us acknowledge the two primary properties of time:

1. A driving rate of change, as fundamental as the $\mathbb{R}^3$ dimensions.
2. A 4th displacement coordinate.

Throughout the following derivations, we presume that time has no physical significance, and acts only as a mathematical tool: $t = \frac{d}{v}$. In turn, we must attribute the 4th displacement coordinate to a position along the density axis that arises, now denoted $y$, and the driving rate of change to *motion*.

Likewise, we presume that in a co-moving frame of reference along the $y$ axis, mechanical equations should limit at their Newtonian counterpart.

As we are removing time as a physically meaningful quantity, let us define

$$\tau = \frac{\dot{x}}{\widehat{v}} \tag{3}$$

where $\dot{x}$ indicates not a derivative with respect to time, but rather a fundamental driving rate of change. Let $\widehat{v} = v \cdot 1t$, or the velocity in units of distance.

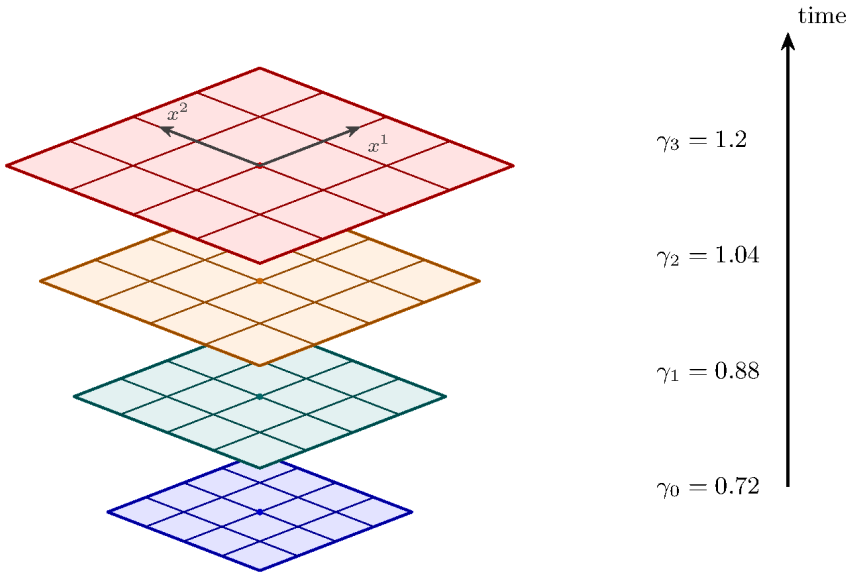



**Fig. 1** A plane-based geometric simplification of the concurrent density axis found by these coordinate transformations.

## 2.1 The Equivalence Principle

Here we examine these coordinate transformations in the context of the equivalence principle. While a well defined function remains elusive, a numerical intersection with the fine structure constant allows us to derive several important properties of an apparent electromagnetic function producing $\vec{g}$.

### 2.1.1 Earth's Kinematic Velocity From Electromagnetic Parameters

By solving $\gamma$ for $v$, let us define $v_{(\gamma)}$ to be

$$v_{(\gamma)} = c\sqrt{1 - \frac{1}{\gamma^2}} \tag{4}$$

And $\delta_{(R)}$ to be

$$\delta_{(R)} = \frac{d}{dR}g = -2\frac{GM}{R^3} \tag{5}$$

If $\delta_{(R)}$ is induced by spatial dilation, we should presume that this dilation is radial (at least in a simplified single body system). We then find the magnitude of a single $\mathbb{R}^3$ axis by taking the inverse of the determinant of a diagonal $3 \times 3$ matrix:

$$\omega = \sqrt[3]{\frac{d}{dR}g} = \delta^{1/3} \tag{6}$$

If we investigate this quantity for Earth, we find:

$$\omega_{\oplus} = 0.014537\ldots \tag{7}$$

While this approximates $2\alpha$, it more closely gives

$$\omega_{\oplus} = \alpha\,(2 - \alpha) \pm 0.0253\% \tag{8}$$

Then by expanding equation 8

$$\alpha\,(2 - \alpha) = \sqrt[3]{\frac{d}{dR}g} \tag{9}$$

Moving the electromagnetic parameters and $R$ to one side before integrating with respect to $R$ gives

$$R^4\left(\frac{e^6\left(\frac{\mu_0}{\epsilon_0}\right)^{\frac{3}{2}}}{4h^3} + \frac{3\mu_0 e^{10}\left(\frac{\mu_0}{\epsilon_0}\right)^{\frac{3}{2}}}{64\epsilon_0 h^5} - \frac{3\mu_0^2 e^8}{16\epsilon_0^2 h^4} - \frac{\mu_0^3 e^{12}}{256\epsilon_0^3 h^6}\right) = 1.272 \times 10^{21} \tag{10}$$

If we then take the radius independent magnitude, isolating the additional electromagnetic parameters we find

$$\frac{e^6\left(\frac{\mu_0}{\epsilon_0}\right)^{\frac{3}{2}}}{4h^3} + \frac{3\mu_0 e^{10}\left(\frac{\mu_0}{\epsilon_0}\right)^{\frac{3}{2}}}{64\epsilon_0 h^5} - \frac{3\mu_0^2 e^8}{16\epsilon_0^2 h^4} - \frac{\mu_0^3 e^{12}}{256\epsilon_0^3 h^6} \tag{11}$$

As $\gamma$ represents 1 plus some small $ds$, let us add 1 to these additional parameters and substitute $\gamma_{\text{em}}$ into equation 4. This gives

$$\gamma_{\text{em}} = 1 + \frac{e^6\left(\frac{\mu_0}{\epsilon_0}\right)^{\frac{3}{2}}}{4h^3} + \frac{3\mu_0 e^{10}\left(\frac{\mu_0}{\epsilon_0}\right)^{\frac{3}{2}}}{64\epsilon_0 h^5} - \frac{3\mu_0^2 e^8}{16\epsilon_0^2 h^4} - \frac{\mu_0^3 e^{12}}{256\epsilon_0^3 h^6} \tag{12}$$

$$v_{(\gamma_{\text{em}})} = 371,721 \ \frac{m}{s} \tag{13}$$

matching the 2018 Plank data[12] to within $\approx 0.52\%$.

### 2.1.2 Solar $\omega$

If we expand equation 9 we find

$$\alpha = \Gamma_{(\sigma)} \left( \frac{d}{dR} g \right)^{1/3} \tag{14}$$

Where $\Gamma_{(\sigma)}$ is some unknown function of spin. While a well defined function remains elusive, we can take advantage of several characteristics for specific local bodies according to table 1 to derive certain characteristics of $\sigma$.

**Table 1** Advantageous properties

| Body | Trait | Value |
|---|---|---|
| Earth | The most mass dense. | 5495 kg $m^{-3}$ |
| Saturn | The least mass dense. | 687 kg $m^{-3}$ |
| Sun | The most massive. | $1.988 \times 10^{30}$ kg |

In equation 9 we've demonstrated that for the most mass dense body in the system

$$\Gamma_{(\sigma),\oplus} \approx \frac{1}{2-\sigma} \tag{15}$$

where $\sigma_{\oplus} \approx \alpha$.

The Sun, being the dominant local mass with a surface gravitational acceleration almost 30 times greater than the next body in question, would project this spatial vector (moving space itself) more closely to a purely radial projection. This behavior, the manner in which the Sun dominates the local gravitational environment should allow us to approximate a single body solution near the solar surface.

Let us then inspect $\Gamma_{(\sigma),\odot}$, under the assumption that this quantity acts *inside* of the divergence, acting on each individual $\mathbb{R}^3$ axis.

$$\Gamma_{(\sigma),\odot} \propto \sqrt[3]{\Gamma_{(\theta)}} \tag{16}$$

If we then presume that $\alpha$ in $\Gamma_{(\theta),\oplus}$ is associated with spin, we can approximate this term to be 0, giving

$$\Gamma_{(\sigma),\odot} \approx \frac{1}{\sqrt[3]{2}} \tag{17}$$

Or

$$\alpha = \Gamma_{(\sigma),\odot} \sqrt[3]{\frac{d}{dR} g} = \sqrt[3]{\frac{1}{2} \frac{d}{dR} g_{\odot}} \pm 0.47\% \tag{18}$$

We find through observation that for the least mass dense body in the system, $\Gamma_{(\sigma),\text{ɦ}} \approx 1$.

This gives a set of 3 equations where

$$\begin{aligned} g_{\oplus} &= R_{\oplus}\left(\frac{\alpha^3\,(2-\alpha)^3}{2}\right) && \pm 0.008\% \\ g_{\odot} &= R_{\odot}\alpha^3 && \pm 1.426\% \\ g_{\text{ɦ}} &= \frac{1}{2}R_{\text{ɦ}}\alpha^3 && \pm 1.134\% \end{aligned} \tag{19}$$

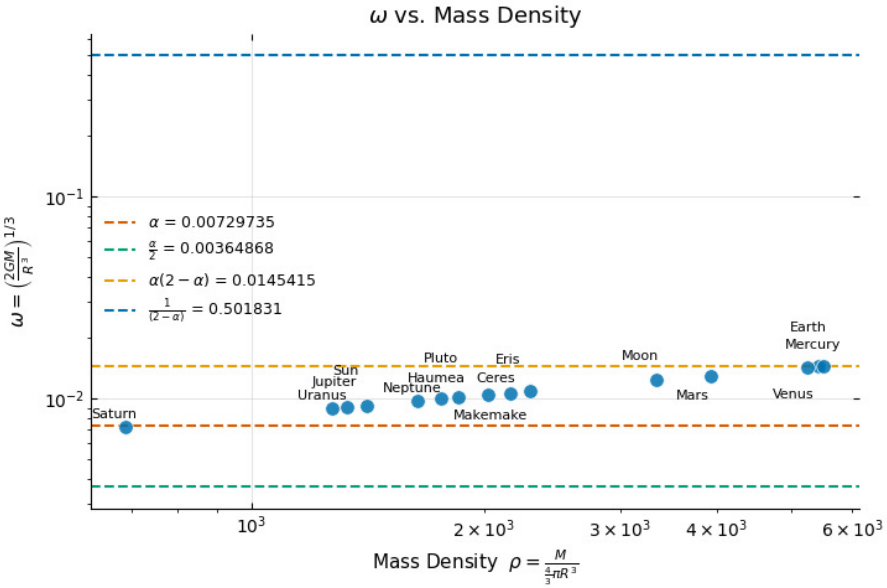



**Fig. 2** The inverse of the determinant representing a change in volume, being equivalent to $dy$, plotted against mass density.

Which in turn produces a domain where $\alpha \lessapprox \omega \lessapprox \alpha\,(2-\alpha)$ as shown in figure 2.

## 2.2 Gravitational Parameters

### 2.2.1 Cosmological Velocity

$$\int_0^R \int_0^x \left[\delta \frac{1}{v_\gamma}\tau\right]\, dR\; d\tau = 1 + \frac{1}{2}\frac{GM_{\oplus}}{R_{\oplus}^3}\left(\frac{\dot{x}}{\hat{x}}\right)^2 \tag{20}$$

### 2.2.2 Kinematic Velocity

$$v_{\text{kinematic}} = c\sqrt{1 - \frac{1}{\left(1 + \frac{1}{2}G\frac{M}{R^3}\right)^2}} = 371,567 \text{ m s}^{-1} \tag{21}$$

## 2.3 Orbital Velocity

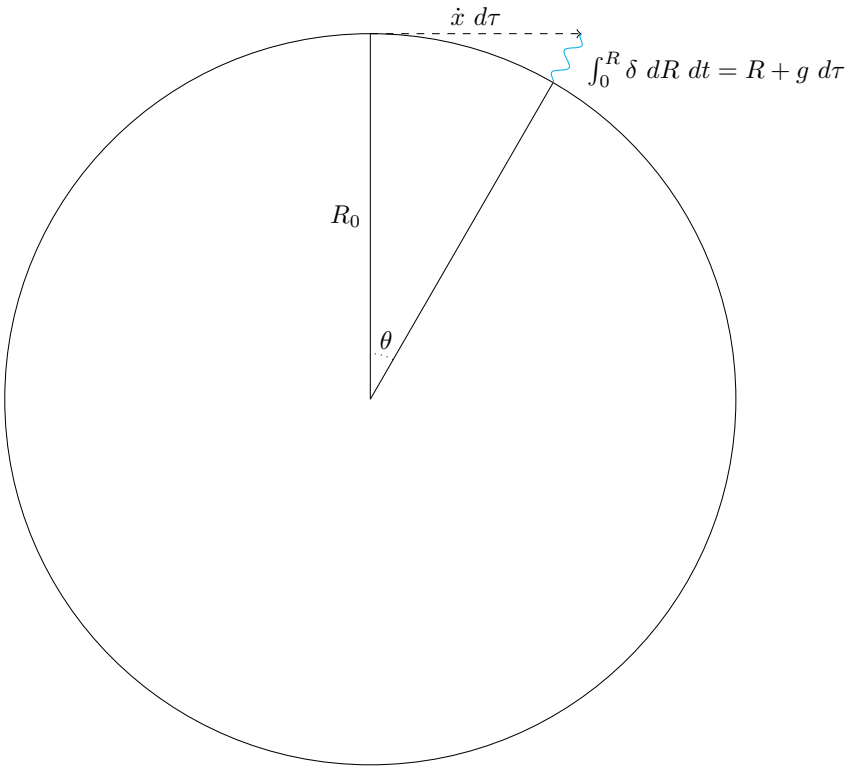



**Fig. 3** A trigonometric derivation of a relativistic orbital velocity equation.

### 2.3.1 Relativistic Stable Orbit Velocity

$$R_0^2 + \dot{x}^2 dt^2 = \left(\int_0^{R_0} 2G\frac{M}{R^3}\right)^2 = (R_0 + g\ dt)^2 \tag{22}$$

$$v_{\text{orbit}} = \frac{\sqrt{GM\left(GM + 2R^3\right)}}{R^2} = 230,145 \text{ m s}^{-1} \tag{23}$$

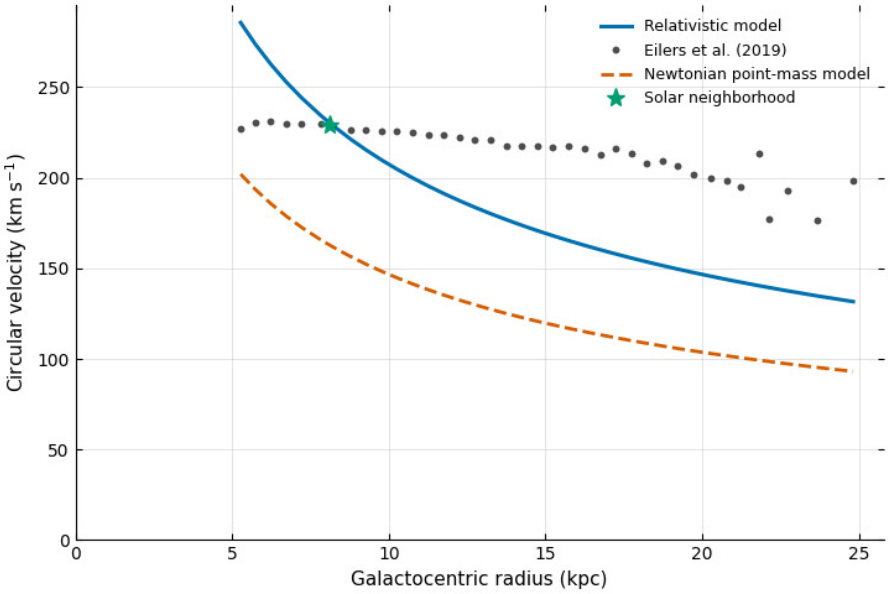



**Fig. 4** Milky Way circular velocity as a function of galactocentric radius. Eilers et al. (2019) measurements are shown as points, alongside the relativistic and Newtonian point-mass model curves. The solar-neighborhood measurement is marked with a star; error bars indicate the reported uncertainties in galactocentric radius and circular velocity.

### 2.3.2 Kinematic Velocity From Triginometric Orbit Equations

$$\mathrm{d}t = \frac{2R^3}{GM - 2R^3} = -1.000000768 \tag{24}$$

$$v_{(-\mathrm{d}t)} = 371,580 \text{ m/s} \tag{25}$$

## 3 Results

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## 5 Equations

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